The Metabolic Module, in Digest

The flux layer, the parameters it is fed,

and what has been measured

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Abstract

The metabolic module is a differentiable layer that maps a perturbed gene representation to a flux vector over yeast-GEM 9.0.2 under stoichiometry, an enzyme-capacity bound and a second-law penalty. Its three parameter tables, turnover numbers, Michaelis constants and formation energies, are meant to be tabulated rather than learned. The tables have been built and audited: six published k_{cat} predictors disagree on a single gene by a median factor of 19, yet 56 to 84 percent of their values land inside the spread of enzymes sharing a function, against 53 percent for a lookup that knows only the function. No trained arm consumes one yet, and the one training grid run so far, 144 cells on Delta, produced constant predictors in every cell, so the module's effect on betaxanthin prediction is unmeasured rather than negative. Full accounts are in the 026 and 027 documents and the flux-layer note.

1 The flux module and what feeds it ✗

The cell graph transformer's instruments map representation to representation (Type I) or to a measured output (Type II). The flux module is neither: it leaves gene index space for reaction space $\mathbb{R}^{4131}$ under a fixed stoichiometry, and its output ν is a latent physical state phenotypes are read from (Figure 1). A deletion sets a gene's availability to zero as a hard fact; the GEM's gene-protein-reaction rules fold gene gates to a reaction by a softmin over a complex's subunits and a sum over isozymes, and the result scales an enzyme-capacity ceiling,

$$|\nu_j| \le c_j(H_{\text{pert}})\, k_{cat,j}\, \bar{E}, \qquad \mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_{\text{th}} \frac{1}{|\mathcal{J}|} \sum_{j \in \mathcal{J}} \frac{[\nu_j\, \Delta_r G_j + \epsilon]_+}{|\nu_j|\,|\Delta_r G_j| + \epsilon}. \tag{1}$$

The capacity bound is the only path by which a genotype reaches a flux magnitude, since 4,129 of the GEM's 4,131 bounds are ± 1000 or 0; $\bar{E}$ is constant and no enzyme abundance is predicted. The second-law term charges for flux running uphill and, because $\Delta_r G$ sums to zero around any cycle, removes with no binary variable the 427 internal loops a growth constraint leaves free. It acts at training time only; the violation fraction is reported at evaluation. Gibbs dissipation is a logged readout, not a penalty: the Niebel 2019 limit is an empirical plateau, not a law.

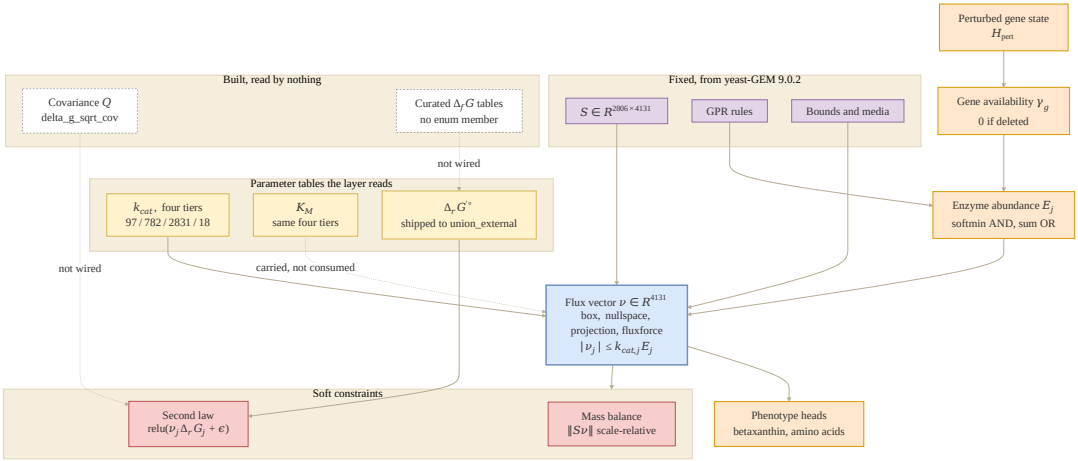


Figure 1: The flux module's main components and what feeds them. Orange, the path from the perturbed gene state to enzyme availability; purple, the fixed yeast-GEM 9.0.2 stoichiometry and bounds; yellow, the parameter tables, a **dashed** border marking one built and read by nothing; blue, the flux vector; red, the soft constraints. Three of the five tables are dashed: k_{cat} resolves from the Open Enzyme Database at 3.7 percent of catalytic units and $\Delta_f G$ from the table shipped with the GEM. A predicted k_{cat} tier and a merged energy route exist in the resolvers, under test. Source `notes/torchcell.models.equivariant_cell_graph_transformer.mermaid.metabolic-module-simple.md`.

2 What the parameter tables are worth ✗

Six published k_{cat} predictors (DLKcat, UniKP, EITLEM, TurNuP, DeepEnzyme, EZKinetics) and four K_M predictors were run from hash-pinned mirrors over 95.3 percent of yeast-GEM's 3,728 catalytic units, the ceiling set by substrate structure availability. Per gene, the unit the capacity bound reads (Figure 2), the median gene moves 1.28 orders of magnitude between the highest and lowest predictor and 71 percent move by more than one order, against a consensus range across genes of 0.87. On the 57 measured yeast pairs, EZKinetics leads on rank at Spearman 0.79 and EITLEM on magnitude at 0.23 orders median error, an upper bound: every predictor trained on the literature those measurements come from.

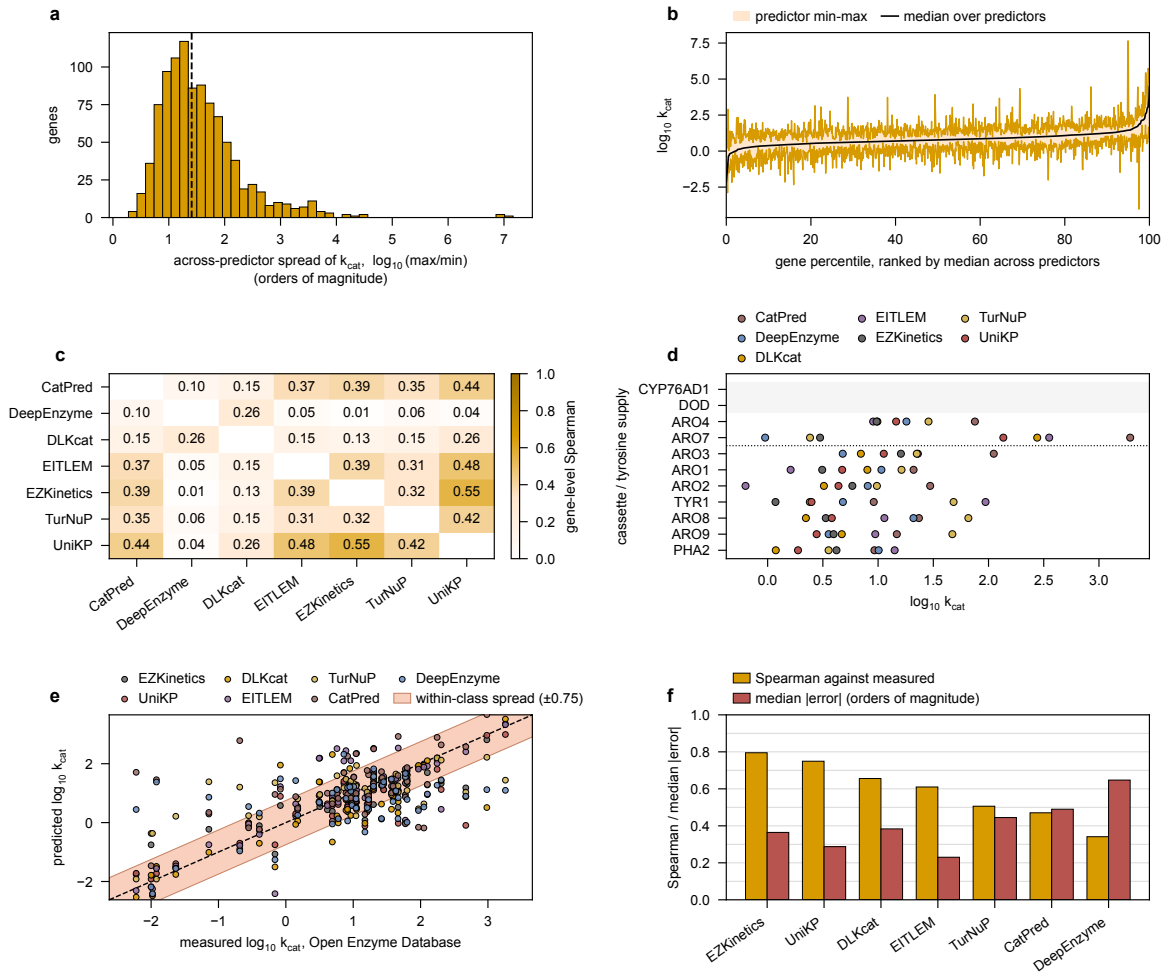


Figure 2: Agreement between the six k_{cat} predictors, and their accuracy against measured yeast values. Panels a to d cover the 980 genes all six predictors cover; e and f the 57 measured pairs. **a**, Across-predictor spread per gene; dashed, the median of 1.28 orders of magnitude. **b**, The same spread against the dynamic range of k_{cat}, genes ranked by their median. **c**, Gene-level Spearman between predictors, 0.01 to 0.55. **d**, The betaxanthin supply genes; the shaded band marks the two heterologous cassette members with no yeast-GEM reaction. **e**, Predicted against measured; the band's half width of 0.75 is the median standard deviation of $\log_{10} k_{cat}$ within one EC4 class. **f**, Spearman and median absolute error, ranked. Figure 6 of the 026 document, from `plot_per_gene_predictor_agreement.py`.

Most of that plausibility is coarse-class knowledge (Figure 4). Against the EC4 band every predictor beats the EC-median lookup, 56 to 84 percent against 53; at EC1 the lookup already reaches 79 percent, and for K_M no predictor beats the lookup at all. On the nine unmeasured betaxanthin supply genes, four of six k_{cat} predictors are class-plausible and no K_M predictor is.

Where the kinetic values come from. The resolved k_{cat} and K_M tables cover all 3,728 catalytic units on four tiers, tried in order: 97 units, 2.6 percent, carry a measurement of yeast's own protein on a substrate the reaction consumes; 782, 21.0 percent, a measurement of the same EC number and substrate in another organism; 2,831, 75.9 percent, the median over the predictors an admission rule keeps, which is EZKinetics, EITLEM and UniKP for k_{cat} and EZKinetics alone for K_M; and 18 the organism default. The mix is therefore about a quarter measured and three quarters estimated, and the homolog tier is counted with the estimates wherever an experimental fraction is quoted, because a measurement of another organism's enzyme is not a measurement of this one. The 18 defaults are a ceiling

rather than a backlog: 14 of the 28 substrate rows behind them carry an R-group placeholder, 7 are macromolecular species with no formula, and the structure writer was run on the rest and refused each.

Where the thermodynamic values come from. The table is recomputed with eQuilibrator component contribution, which returns a covariance the shipped point values cannot, and it now prices **3,801 of 4,131 reactions**, 3,535 of the 3,847 chemical ones and **2,724 of the 2,845 that can carry flux**. Every value names one of nine routes and every gap one of seven failure classes (Figure 3). The mix is dominated by one evidence class: 2,494 of the 2,559 priced species are estimated by component contribution, 59 read from a table, 4 fixed by a curation record and 2 gauged from a measured midpoint potential. **Coverage is not what the eight recovery routes past the plain accession lookup bought**; between them they price 153 species. What they bought is a table in which every value names its own method.

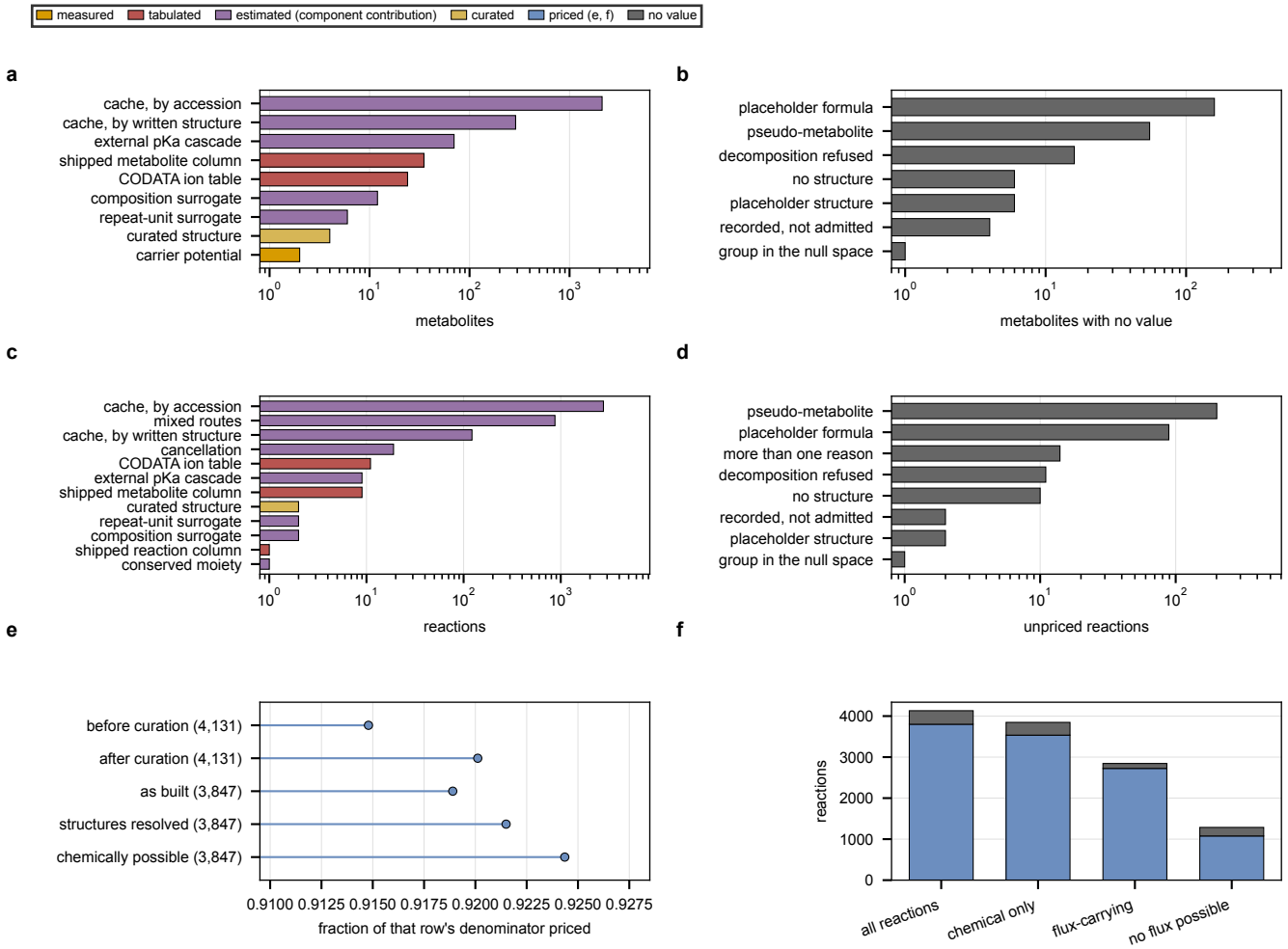


Figure 3: Which route produced each formation and reaction energy, and which failure class holds the rest. Bars are colored by evidence class; panels a to d use a logarithmic count axis. **a**, The 2,559 priced metabolites by route. **b**, The 247 with no value, by failure class. **c**, The 3,801 priced reactions by route; *mixed routes* is a reaction whose participants came from more than one. **d**, The 330 unpriced reactions by the single class that blocks them. **e**, The ceiling ladder, each row against the denominator in its own tick label. The last rung, 3,556 of 3,847 chemical reactions, is what would be priced if every structure, ladder and refused decomposition were resolved, so it is the ceiling for any method rather than for this one. **f**, Reactions priced under four denominators. Figure 11 of the 026 document, from `plot_thermo_recovery_routes.py`.

The shipped file is in kilocalories and labeled kilojoules, copied verbatim from yETFL by yeast-GEM's curation script: regressing our table on yETFL's over 1,927 shared metabolites gives a slope of 4.187 at $r = 0.987$. Correcting that and matching compartment conditions closes the transport arm of the disagreement, median 17.4 to 5.6 kJ mol^{-1}, leaving the single-compartment cloud at 17.6, the genuine estimator difference.

Two recovery routes are worth naming because they are error sources as much as coverage. Structures written from the metabolite's own name price 347 lipid rows, and 48 of them need a compound eQuilibrator has never seen, created through a pK_a cascade selected against the compound cache's own constants on 2,100 held-out compounds. That cascade's ladder error runs to a 90th percentile of 8.33 kJ mol^{-1} at the worst compartment pH, while writing the same structure as its neutral parent rather than its pH 7 major microspecies moves the answer by a median of 1,585 kJ mol^{-1}, so the reference convention and not the predictor is what has to be got right. A composition check over the

2,414 accession-priced species also found 8 priced as a different molecule, 6 of which are withdrawn.

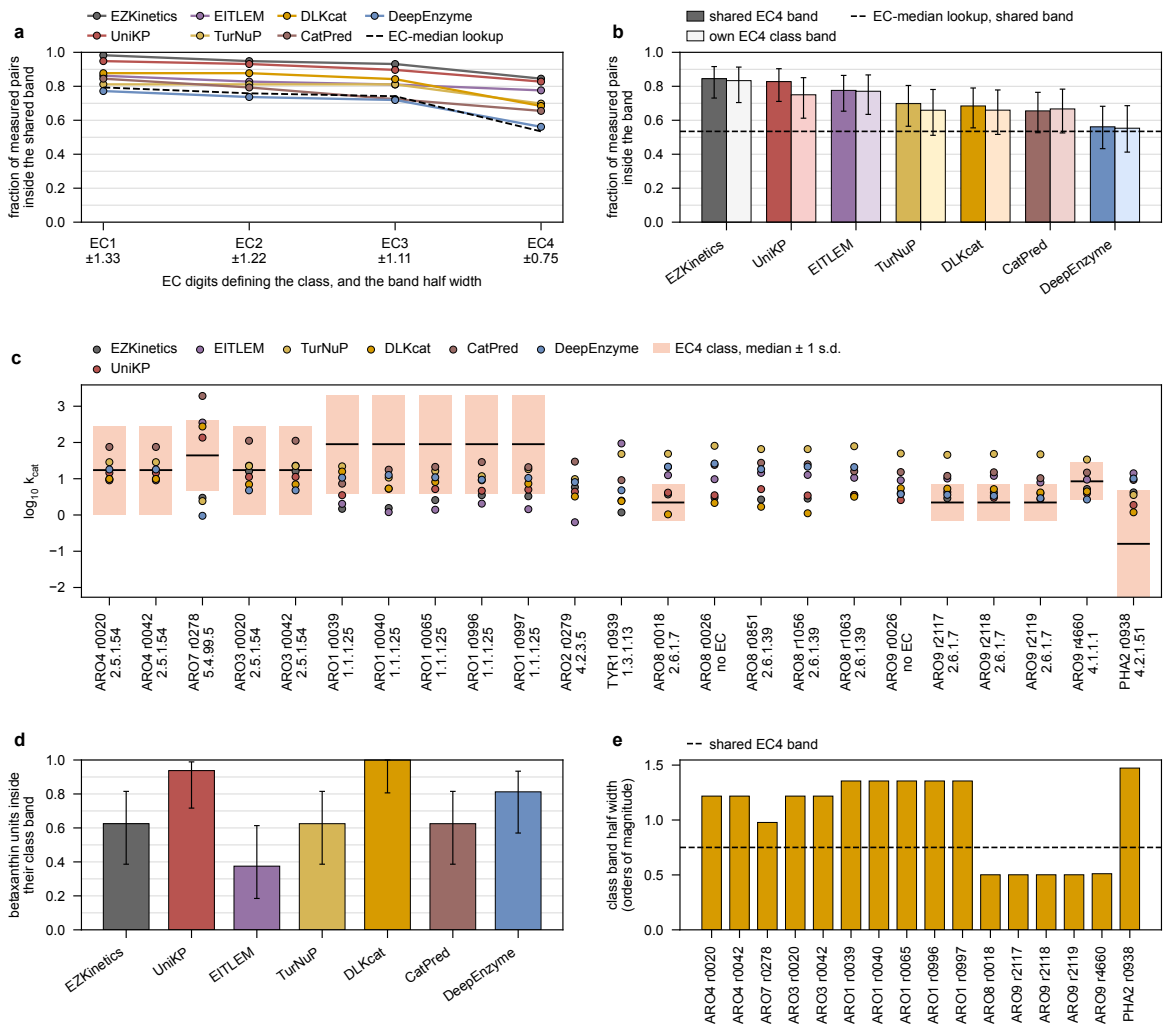


Figure 4: How many of each predictor's k_{cat} values fall inside the spread of enzymes doing the same job. a, Fraction of the 57 measured pairs inside a shared band as the class is read to fewer EC digits; dashed, the EC-median lookup. **b**, The EC4 fraction against each pair's own class band, Wilson 95 percent intervals. **c**, The 23 betaxanthin supply units against one standard deviation of their EC4 class. **d**, Fraction of the 16 scorable units inside, by predictor: DLKcat 16 of 16, UniKP 15, EITLEM 6. **e**, Each unit's class band half width. Figure 7 of the 026 document, from `plot_within_class_plausibility.py`.

3 Where the training stands ✗

The 027 grid on Delta ran 144 cells, six arms over 24 seeds, on Cachera's betaxanthin screen with the 019 nine-graph backbone. The anchored module scored −0.020 test Spearman against +0.027 for no module. The prediction dumps show every cell is a constant predictor: the largest prediction spread in any cell is 0.8 percent of the target spread, the no-module arms sit at the target mean with a training loss of exactly 1.00 from epoch 10, and the flux arms at wrong constants. The claim that the thermodynamic anchor costs is therefore withdrawn and the module's effect is unmeasured; the same backbone reached 0.434 under the 019 trainer, so the defect is in the 027 loop. Two defects precede any relaunch: training uses yeast-GEM's shipped exchange bounds, which forbid all amino-acid uptake, while the Cachera strains grew in a supplemented medium; and 170 of the 183 proton-translocating reactions carry a confidently wrong zero, the translocation term being computed but off by default.

Round 2 has one positive result, and it is not on betaxanthin. Supervising the flux layer's own learned whole-cell concentration, the volume-weighted mixture over the compartments each amino acid occupies, as the amino-acid readout beats the pooled control at all three seeds: +0.0145 mean per-feature Spearman on the pinned test split, standard deviation 0.0093, at an epoch selected on validation and never a maximum over epochs, and +0.0198 (s.d. 0.0136) with the constraint scale at 0. Wave 6's turnover readout on the byte-identical arm collapses to a constant, −0.0494 against the same control at its two seeds, so concentration minus turnover is +0.0686 at matched seeds. The learned whole-cell medians match the measured medians to 0.036 to 0.037 decades over four orders of magnitude, while the per-column held-out Spearman runs only −0.09 to +0.17 and the unsupervised arm reads −0.0031, so the agreement

is supervision rather than physics. Four caveats travel with it: 10.63 percent of Mulleder's measurements sat above the shipped 10^{-2} M ceiling, so the window was widened to 1 M before the round; five of the six fractional volumes are assumed rather than sourced; $n = 3$ carries two degrees of freedom; and betaxanthin parity on the one shipped kinetic table is -0.055 (s.d. 0.090, $n = 3$), a null rather than a cost. Full account, contrasts and caveats in the 027 document.

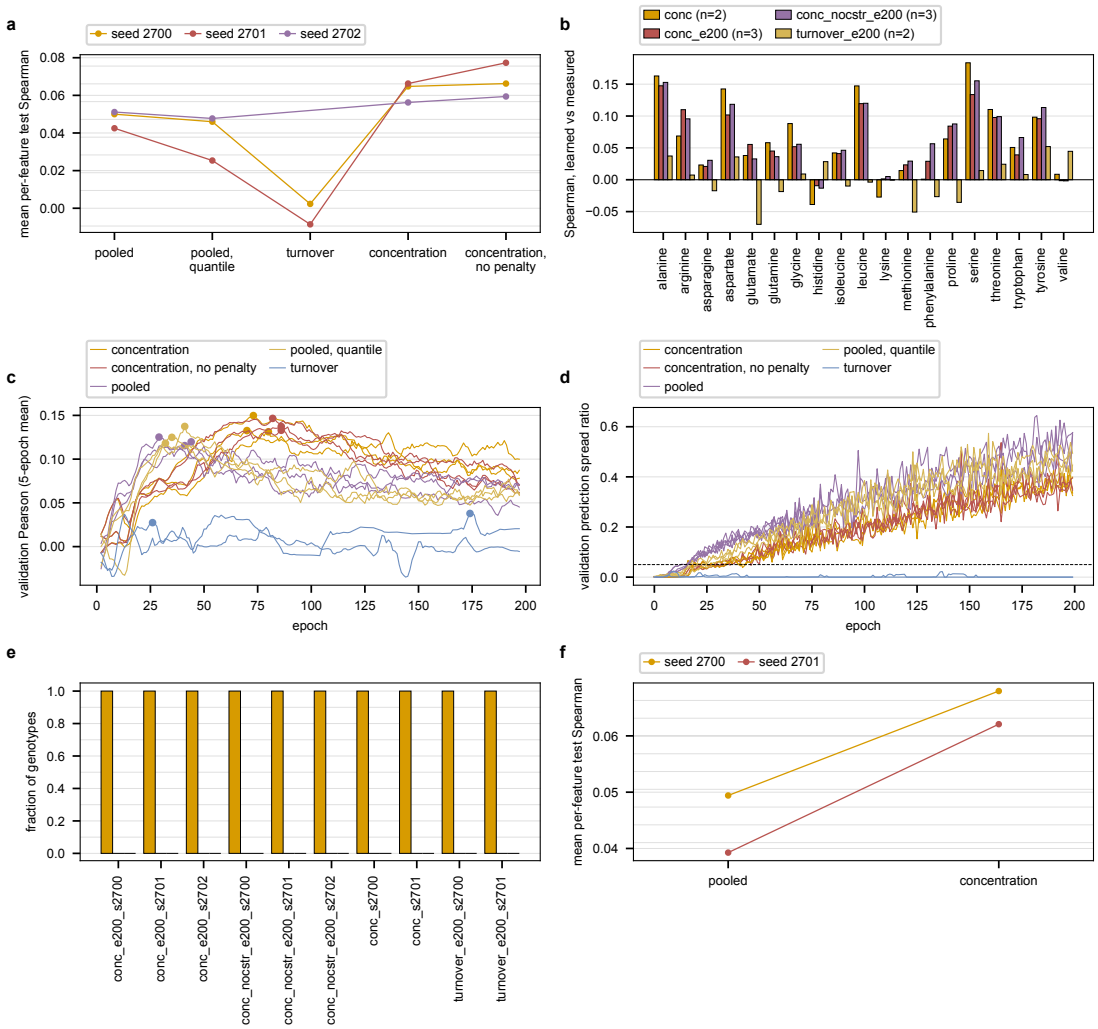


Figure 5: The amino-acid concentration readout against the pooled control. **a**, Mean per-feature pinned-test Spearman per readout, one line per seed, so a slope is the paired difference. **b**, The same statistic per amino acid, learned whole-cell concentration against Mulleder's measurement, by arm. **c**, Validation Pearson as a centered five-epoch mean, one line per cell colored by its arm, with the selected epoch marked. **d**, Validation prediction spread ratio against the 0.05 collapse line; the two turnover cells sit on it. **e**, Three component feasibility criteria per flux cell: mass balance within 1 percent holds everywhere, the flux box and the no-uphill criterion nowhere, drawn as a flat mark at zero. **f**, The same contrast at the 80-epoch budget, two seeds. Figure 5 of the 027 document, from `analyze_wave7_aa.py`.

What each target can support, measured from each source's own repeats. A model cannot correlate with a measurement better than the measurement correlates with itself, so the square root of a target's reliability is the highest Pearson correlation any predictor of the true level can reach, and it is what a flat result has to be read against. Six metabolite datasets enter the build, three of them decoded by a trained head (Table 1). Betaxanthin is the strongest of the three at a reliability of 0.84 and a ceiling of 0.92, because the CRI-SPA release averages a median of 15 colonies per gene; Cachera's own trial-to-trial correlation of 0.765, carried to four screens by the Spearman-Brown relation, agrees at 0.93. The flat betaxanthin results are therefore a property of the predictor and not of the label. The amino acids sit at 0.42 and 0.62, and the isobutanol screen at 0.11 and 0.33, which reverses the earlier reading that called isobutanol the best-conditioned target: its widely quoted retest of 0.92 was computed on the re-screened strong hits alone, and selecting both tails inflates it. Of the three context datasets, the lipids and the metabolome could support a head at 0.91 and 0.79, while the carotenoid screen's six-level visual score reaches only 0.33. The fourth endpoint, essentiality, is bounded instead by how far two curated sources agree, which is 98.8 percent of 1,121 genes against a majority rate of 86.0 percent.

The parameter tables are closed and the trainer is not. As of 2026-09-11 the formation energies stand at 3,801 of 4,131 reactions and 2,724 of the 2,845 that can carry flux, and the kinetic tables at 97 measured, 782 homolog, 2,831

Table 1: Reproducibility ceiling of every metabolically relevant target in the 027 build, each from its own source's repeats. Reliability is the fraction of the variance in the stored value that is not measurement noise, and its square root is the highest Pearson correlation any predictor of the true level can reach against that stored value. The estimator differs by what each release ships: a variance split where it gives a mean, a standard deviation and a count; a one-way intraclass correlation where it gives repeated raw measurements; a retest correlation where it gives exactly two; and a back-solve from the released dispersion where it gives no replicate at all. Features is the number of measured quantities the reliability is a median over. The lower block enters the build as genotype context, with no head decoding it.

Target	Estimator	Repeats	Features	Reliability	Ceiling
Read by a trained head					
Betaxanthin	variance components	4696	1	0.84	0.92
Amino acids	retest correlation	194	19	0.42	0.62
Isobutanol biosensor	retest correlation	190	1	0.11	0.33
Present as genotype context only					
Carotenoid	one-way ICC	129	1	0.11	0.33
Metabolome (Zelezniak)	one-way ICC, median over metabolites	18	43	0.62	0.79
Lipids (da Silveira)	dispersion back-solve from the released SSMD	130	138	0.82	0.91

predicted and 18 default over 3,728 catalytic units, with every value carrying a named route and every gap a named failure class (Section 2). 291 of the 330 unpriced reactions are permanently closed, blocked only by species that are not molecules, so no further method moves them. Nothing in the parameter tables is now what holds the module back.

A third defect precedes a relaunch, and it is now a config value rather than missing code. Every 026 and 027 run so far read `metabolite_dgf_prime_with_external.parquet`, which predates the composition check and therefore still carries the glycogen price taken from a four-residue oligomer and the dolichol offset. `ThermoSource` now carries a `verified` member that resolves the curated tables through their manifest's sha256 and raises rather than falling back when a hash does not match, and it is the only route in which the composition guard has run. No arm asks for it yet.

Order of work. A trainer that reproduces 0.434 on the no-module arm with an abort on prediction collapse; the module arms on it, on the dataset's actual medium, reading the verified tables; then the merged $\Delta_f G$ and predicted k_{cat} tables, on GilaHyper jobs 1663, 1667, 1668 and 1669, carried to a selected epoch.